Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Carvalho Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
11443 Crows Landing RD	Crows Landing	Stanislaus	95313

Street and nearest cross street (if no address): Crows Landing And Simmons

Date facility was originally placed in operation: 06/01/1952

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X022-X023-XXXX X058-X022-X027-XXXX

B. OPERATORS

Carvalho, Fran

Operator name: Carvalho, Fran

Telephone no.: (209) 632-1406 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11443 Crows Landing RD	Crows Landing	CA	95313

Carvalho, Michael

Operator name: Carvalho, Michael

Telephone no.: (209) 765-7354 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11443 Crows Landing RD	Crows Landing	CA	95313

This operator is responsible for paying permit fees.

C. OWNERS

Carvalho, Fran

Legal owner name: Carvalho, Fran

Telephone no.: (209) 632-1406 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11443 Crows Landing RD	Crows Landing	CA	95313

Carvalho, Michael

Legal owner name:	Carvalho, Michael	Telephone no.:		(209) 765-7354
			Landline	Cellular
11443 Crows Landing RD		Crows Landing	CA	95313
Mailing Address Number and Street		City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	300	300	100	0
Number under roof	600	100	0	0	0	0
Maximum number	600	100	300	300	100	0
Average number	600	100	300	300	100	0
Avg live weight (lbs)	1,400	1,400	900	650		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,148.98 *tons per reporting period*

Total nitrogen from manure: 283,369.41 *lbs per reporting period*

After ammonia losses (30% loss applied): 198,358.59 *lbs per reporting period*

Total phosphorus from manure: 46,811.62 *lbs per reporting period*

Total potassium from manure: 115,693.94 *lbs per reporting period*

Total salt from manure: 305,505.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 32,076,000 *gallons*

Total nitrogen generated: 74,031.90 *lbs*

Total phosphorus generated: 15,543.73 *lbs*

Total potassium generated: 204,401.56 *lbs*

Total salt generated: 1,376,560.49 *lbs*

	32,076,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	32,076,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Lateral 5	Ground water
TID Lateral 5 1/2	Ground water
TID Lateral 6	Ground water
TID Lateral 7	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	55	55	2	process wastewater	X058-X022-X027-XXXX
Field 2	40	40	2	process wastewater	X058-X021-X007-XXXX
Field 3	40	40	2	process wastewater	X058-X021-X011-XXXX
Field 4	16	16	2	process wastewater	X058-X022-X014-XXXX
Field 5	18	12	2	manure	X058-X021-X003-XXXX
Field 6	40	40	2	process wastewater	X058-X022-X023-XXXX
Fulkerth	40	40	0	none	X022-X022-X007-XXXX
Totals for areas that were used for application	209	203	12		
Totals for areas that were not used for application	40	40	0		
Land application area totals	249	243	12		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/30/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 55 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	870.00 *ton*	As-is		76.6	4,380.00	810.00	6,140.00		9.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	15.82	138.57	25.63	194.25	722.52

Field 1

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,332.00 *ton*	As-is		68.6	3,490.00	420.00	5,220.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.22	169.04	20.34	252.84	1,003.80

Field 2

Field name: Field 2

10/31/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 40 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	652.00 *ton*	As-is		61.2	4,680.00	1,080.00	7,050.00		8.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	16.30	152.57	35.21	229.83	1,035.94

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	971.00 *ton*	As-is		65.0	3,990.00	530.00	7,490.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.28	193.71	25.73	363.64	1,036.54

Field 3

Field name: Field 3

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	637.00 *ton*	As-is		56.4	4,920.00	920.00	6,340.00		6.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	15.93	156.70	29.30	201.93	909.57

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	955.00 *ton*	As-is		62.4	4,170.00	490.00	6,160.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.88	199.12	23.40	294.14	1,184.96

Field 4

Field name: Field 4

10/29/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 16 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	253.00 *ton*	As-is		68.2	4,110.00	850.00	5,960.00		8.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	15.81	129.98	26.88	188.49	868.90

Field 4

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	390.00 *ton*	As-is		68.8	3,870.00	570.00	6,160.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.38	188.66	27.79	300.30	1,125.54

Field 5

Field name: Field 5

10/28/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 12 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	203.00 *ton*	As-is		55.0	4,410.00	960.00	5,750.00		6.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	16.92	149.21	32.48	194.54	1,042.91

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	304.00 *ton*	As-is		70.8	2,830.00	570.00	4,300.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.33	143.39	28.88	217.87	976.45

Field 6

Field name: Field 6

10/29/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 40 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	656.00 *ton*	As-is		55.0	4,410.00	960.00	5,750.00		6.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	16.40	144.65	31.49	188.60	1,011.06

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	940.00 *ton*	As-is		68.0	4,100.00	470.00	8,190.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.50	192.70	22.09	384.93	1,052.80

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/30/2022: Triticale, soft dough

Field name: Field 1

Crop: Triticale, soft dough — Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.06	13.98	149.64	751.16	1,296,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	219.37	6,885,000.00 *gal*
Application event totals		55.06	13.98	149.64	970.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.32	16.66	193.89	869.14	1,296,000.00 *gal*
Application event totals		63.32	16.66	193.89	869.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.76	13.88	161.57	724.29	1,080,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	193.57	6,075,000.00 *gal*
Application event totals		52.76	13.88	161.57	917.85	

Field 1 - 05/28/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 1 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.32	10.30	162.82	1,467.36	1,656,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	296.80	9,315,000.00 *gal*
Application event totals		63.32	10.30	162.82	1,764.16	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	232.28	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	232.28	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.05	7.17	113.26	1,020.77	1,152,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	206.47	6,480,000.00 *gal*
Application event totals		44.05	7.17	113.26	1,227.24	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.05	7.17	113.26	1,020.77	1,152,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	206.47	6,480,000.00 *gal*
Application event totals		44.05	7.17	113.26	1,227.24	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.54	6.27	99.11	893.18	1,008,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	180.66	5,670,000.00 *gal*
Application event totals		38.54	6.27	99.11	1,073.84	

Field 1 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	180.66	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	180.66	

Field 2 - 10/31/2022: Triticale, soft dough

Field name: Field 2

Crop: Triticale, soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.47	12.82	137.17	688.56	864,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	212.92	4,860,000.00 *gal*
Application event totals		50.47	12.82	137.17	901.49	

01/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.39	20.36	236.97	1,062.29	1,152,000.00 *gal*
Application event totals		77.39	20.36	236.97	1,062.29	

03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.71	17.81	207.35	929.50	1,008,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	248.41	5,670,000.00 *gal*
Application event totals		67.71	17.81	207.35	1,177.91	

Field 2 - 05/30/2023: Corn, silage

Field 2 - 05/30/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.14	11.09	175.20	1,579.01	1,296,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	319.38	7,290,000.00 *gal*
Application event totals		68.14	11.09	175.20	1,898.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	283.90	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	283.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.21	8.01	126.54	1,140.39	936,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	230.67	5,265,000.00 *gal*
Application event totals		49.21	8.01	126.54	1,371.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.99	8.62	136.27	1,228.12	1,008,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	248.41	5,670,000.00 *gal*
Application event totals		52.99	8.62	136.27	1,476.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.42	7.39	116.80	1,052.67	864,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	212.92	4,860,000.00 *gal*
Application event totals		45.42	7.39	116.80	1,265.59	

Field 2 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	177.44	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	177.44	

Field 3 - 10/28/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2021	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.68	13.88	148.60	745.94	936,000.00 *gal*
Application event totals		54.68	13.88	148.60	745.94	

10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	230.67	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	230.67	

02/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.39	20.36	236.97	1,062.29	1,152,000.00 *gal*
Application event totals		77.39	20.36	236.97	1,062.29	

03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.71	17.81	207.35	929.50	1,008,000.00 *gal*
Application event totals		67.71	17.81	207.35	929.50	

Field 3 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	212.92	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	212.92	

Field 3 - 05/30/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.56	9.85	155.74	1,403.56	1,152,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	283.90	6,480,000.00 *gal*
Application event totals		60.56	9.85	155.74	1,687.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	248.41	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	248.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.64	6.77	107.07	964.95	792,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	195.18	4,455,000.00 *gal*
Application event totals		41.64	6.77	107.07	1,160.13	

Field 3 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.21	8.01	126.54	1,140.39	936,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	230.67	5,265,000.00 *gal*
Application event totals		49.21	8.01	126.54	1,371.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.42	7.39	116.80	1,052.67	864,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	212.92	4,860,000.00 *gal*
Application event totals		45.42	7.39	116.80	1,265.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	195.18	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.18	

Field 4 - 10/29/2022: Triticale, soft dough

Field name: Field 4

Crop: Triticale, soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.57	13.35	142.89	717.25	360,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	221.79	2,025,000.00 *gal*
Application event totals		52.57	13.35	142.89	939.05	

Field 4 - 10/29/2022: Triticale, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.55	19.08	222.16	995.89	432,000.00 *gal*
Application event totals		72.55	19.08	222.16	995.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.46	15.90	185.13	829.91	360,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	221.79	2,025,000.00 *gal*
Application event totals		60.46	15.90	185.13	1,051.70	

Field 4 - 05/31/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.24	10.78	170.34	1,535.15	504,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	310.51	2,835,000.00 *gal*
Application event totals		66.24	10.78	170.34	1,845.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	266.15	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	266.15	

Field 4 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.58	6.93	109.50	986.88	324,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	199.62	1,822,500.00 *gal*
Application event totals		42.58	6.93	109.50	1,186.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.32	7.70	121.67	1,096.53	360,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	221.79	2,025,000.00 *gal*
Application event totals		47.32	7.70	121.67	1,318.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.85	6.16	97.34	877.23	288,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	177.44	1,620,000.00 *gal*
Application event totals		37.85	6.16	97.34	1,054.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	177.44	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	177.44	

Field 5 - 10/28/2022: Triticale, soft dough

Field name: Field 5

Crop: Triticale, soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 5 - 10/28/2022: Triticale, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	192.91	74.90	74.90	2,767.88	275.00 *ton*
Application event totals		192.91	74.90	74.90	2,767.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	236.58	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	236.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	236.58	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	236.58	

Field 5 - 05/31/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	196.35	35.23	124.74	13,686.75	300.00 *ton*
Application event totals		196.35	35.23	124.74	13,686.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	295.73	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	295.73	

Field 5 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	266.15	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	266.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	236.58	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	236.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	236.58	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	236.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	236.58	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	236.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	207.01	1,417,500.00 *gal*
Application event totals		0.00	0.00	0.00	207.01	

Field 6 - 10/29/2022: Triticale, soft dough

Field name: Field 6

Crop: Triticale, soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 6 - 10/29/2022: Triticale, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.68	13.88	148.60	745.94	936,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	230.67	5,265,000.00 *gal*
Application event totals		54.68	13.88	148.60	976.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.39	20.36	236.97	1,062.29	1,152,000.00 *gal*
Application event totals		77.39	20.36	236.97	1,062.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.04	15.27	177.73	796.71	864,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	212.92	4,860,000.00 *gal*
Application event totals		58.04	15.27	177.73	1,009.64	

Field 6 - 05/29/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.35	10.47	165.47	1,491.28	1,224,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	301.64	6,885,000.00 *gal*
Application event totals		64.35	10.47	165.47	1,792.93	

Field 6 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	248.41	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	248.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.42	7.39	116.80	1,052.67	864,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	212.92	4,860,000.00 *gal*
Application event totals		45.42	7.39	116.80	1,265.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.21	8.01	126.54	1,140.39	936,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	230.67	5,265,000.00 *gal*
Application event totals		49.21	8.01	126.54	1,371.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.42	7.39	116.80	1,052.67	864,000.00 *gal*
TID Lateral 5	Ground water	0.00	0.00	0.00	212.92	4,860,000.00 *gal*
Application event totals		45.42	7.39	116.80	1,265.59	

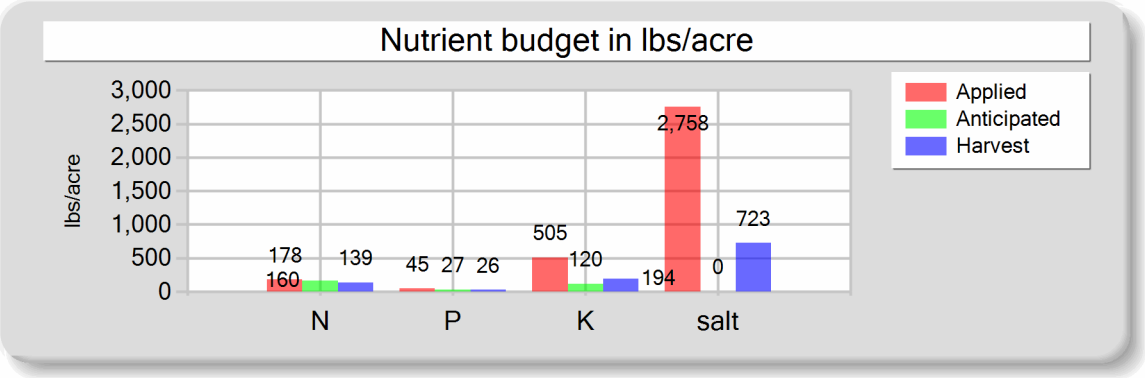
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 5	Ground water	0.00	0.00	0.00	195.18	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.18	

B. NUTRIENT BUDGET

Field 1 - 10/30/2022: Triticale, soft dough

Field name: Field 1 Crop: Triticale, soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.14	44.52	505.10	2,344.59
Fresh water	0.00	0.00	0.00	412.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.14	44.52	505.10	2,757.53
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	138.57	25.63	194.25	722.52
Nutrient balance	39.57	18.89	310.85	2,035.00
Applied to removed ratio	1.29	1.74	2.60	3.82

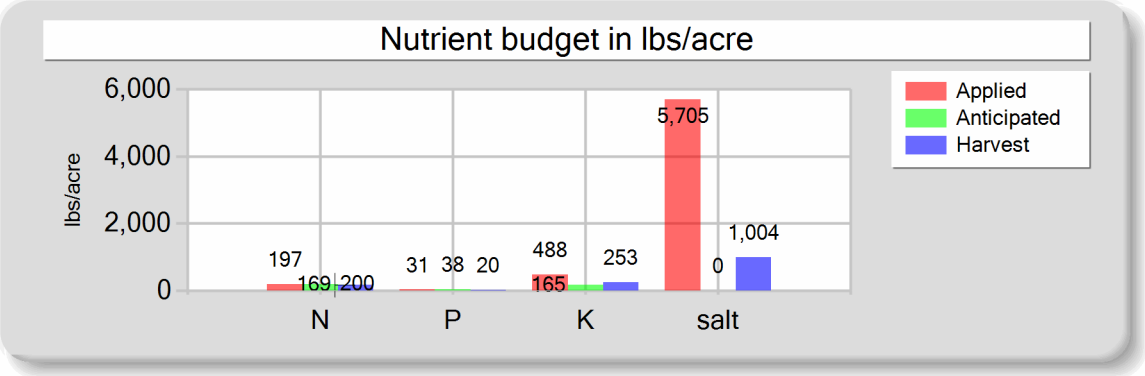
Fresh water applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
8.68	*inches/acre*

Process wastewater applied	
3,672,000.00	*gallons*
135.23	*acre-inches*
2.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/28/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	189.95	30.91	488.45	4,402.08
Fresh water	0.00	0.00	0.00	1,303.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.95	30.91	488.45	5,705.43
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	169.04	20.34	252.84	1,003.80
Nutrient balance	27.91	10.56	235.61	4,701.63
Applied to removed ratio	1.17	1.52	1.93	5.68

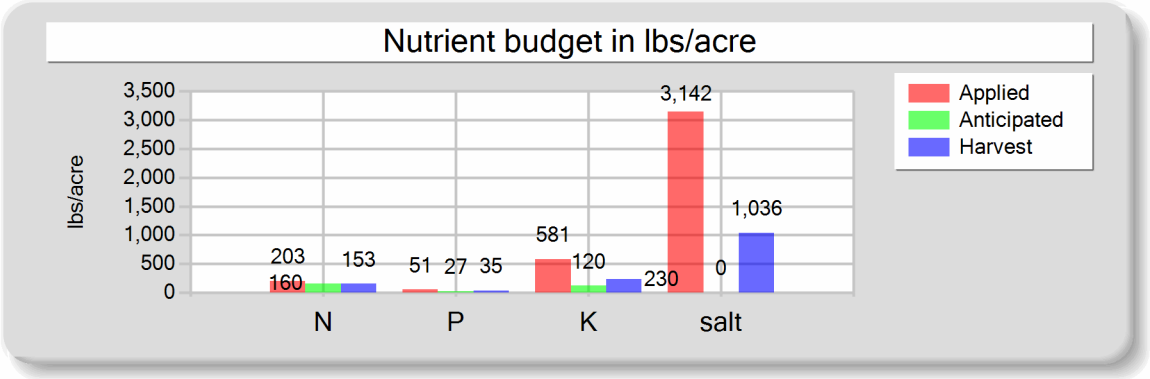
Fresh water applied	
40,905,000.00	*gallons*
1,506.39	*acre-inches*
27.39	*inches/acre*

Process wastewater applied	
4,968,000.00	*gallons*
182.95	*acre-inches*
3.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/31/2022: Triticale, soft dough

Field name: Field 2 Crop: Triticale, soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	195.57	50.98	581.49	2,680.35
Fresh water	0.00	0.00	0.00	461.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.57	50.98	581.49	3,141.68
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	152.57	35.21	229.83	1,035.94
Nutrient balance	50.01	15.78	351.66	2,105.74
Applied to removed ratio	1.33	1.45	2.53	3.03

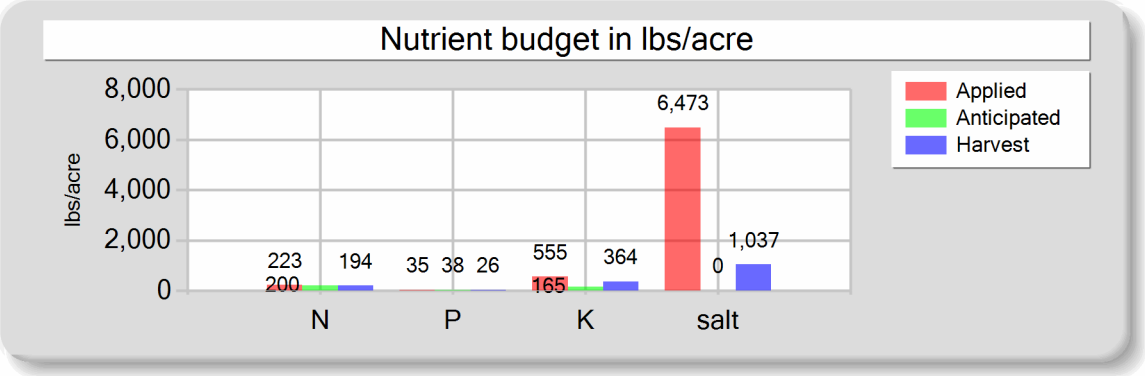
Fresh water applied	
10,530,000.00	*gallons*
387.78	*acre-inches*
9.69	*inches/acre*

Process wastewater applied	
3,024,000.00	*gallons*
111.36	*acre-inches*
2.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/30/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.76	35.10	554.82	5,000.19
Fresh water	0.00	0.00	0.00	1,472.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.76	35.10	554.82	6,472.91
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	193.71	25.73	363.64	1,036.54
Nutrient balance	29.05	9.37	191.18	5,436.36
Applied to removed ratio	1.15	1.36	1.53	6.24

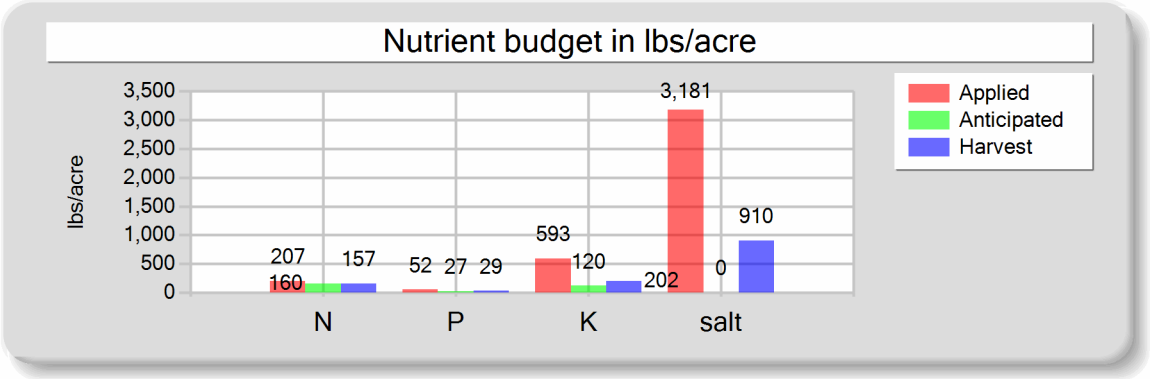
Fresh water applied	
33,615,000.00	*gallons*
1,237.93	*acre-inches*
30.95	*inches/acre*

Process wastewater applied	
4,104,000.00	*gallons*
151.14	*acre-inches*
3.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/28/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.78	52.05	592.92	2,737.73
Fresh water	0.00	0.00	0.00	443.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.78	52.05	592.92	3,181.32
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	156.70	29.30	201.93	909.57
Nutrient balance	50.08	22.75	390.99	2,271.74
Applied to removed ratio	1.32	1.78	2.94	3.50

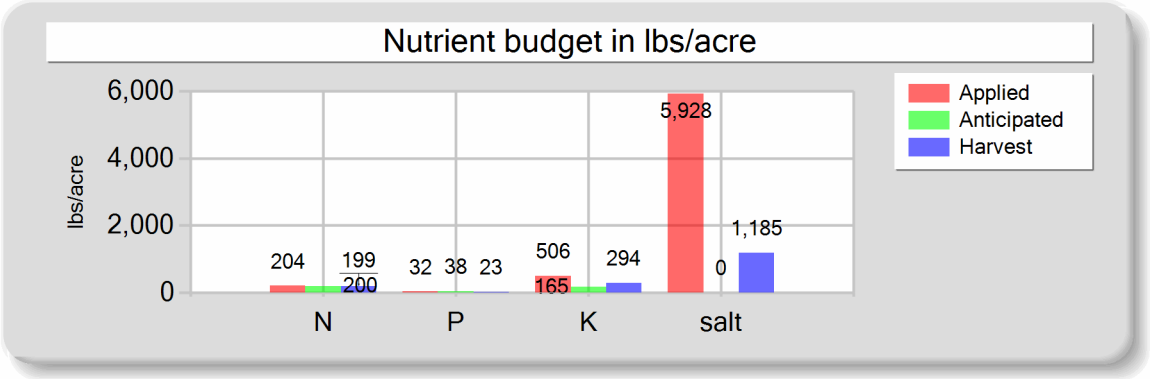
Fresh water applied	
10,125,000.00	*gallons*
372.87	*acre-inches*
9.32	*inches/acre*

Process wastewater applied	
3,096,000.00	*gallons*
114.02	*acre-inches*
2.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/30/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.84	32.02	506.15	4,561.58
Fresh water	0.00	0.00	0.00	1,366.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.84	32.02	506.15	5,927.83
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	199.12	23.40	294.14	1,184.96
Nutrient balance	4.72	8.63	212.01	4,742.87
Applied to removed ratio	1.02	1.37	1.72	5.00

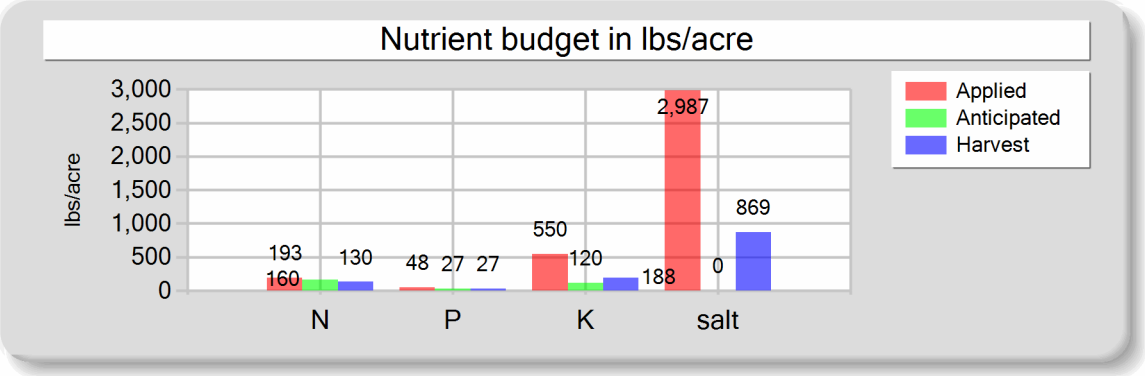
Fresh water applied	
31,185,000.00	*gallons*
1,148.44	*acre-inches*
28.71	*inches/acre*

Process wastewater applied	
3,744,000.00	*gallons*
137.88	*acre-inches*
3.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 10/29/2022: Triticale, soft dough

Field name: Field 4 Crop: Triticale, soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	185.58	48.34	550.18	2,543.06
Fresh water	0.00	0.00	0.00	443.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.58	48.34	550.18	2,986.64
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	129.98	26.88	188.49	868.90
Nutrient balance	62.61	21.46	361.70	2,117.74
Applied to removed ratio	1.48	1.80	2.92	3.44

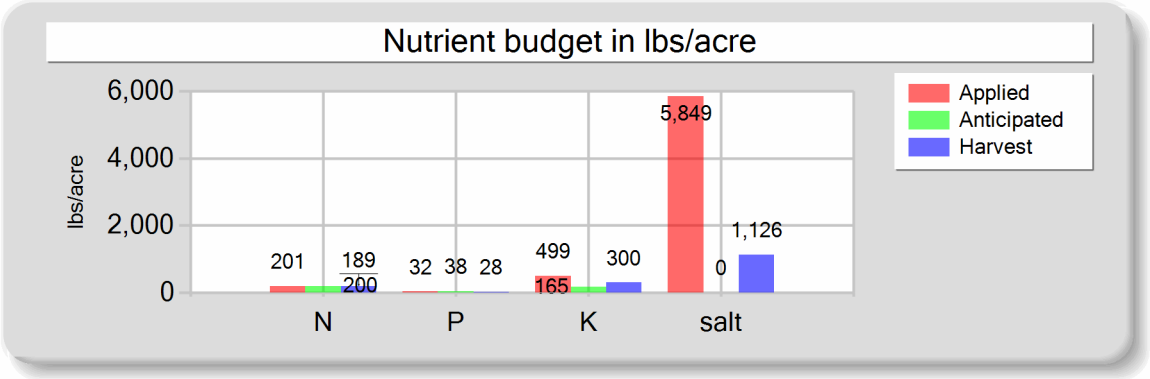
Fresh water applied	
4,050,000.00	*gallons*
149.15	*acre-inches*
9.32	*inches/acre*

Process wastewater applied	
1,152,000.00	*gallons*
42.42	*acre-inches*
2.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 05/31/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.00	31.56	498.85	4,495.79
Fresh water	0.00	0.00	0.00	1,352.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.00	31.56	498.85	5,848.73
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	188.66	27.79	300.30	1,125.54
Nutrient balance	12.33	3.78	198.55	4,723.19
Applied to removed ratio	1.07	1.14	1.66	5.20

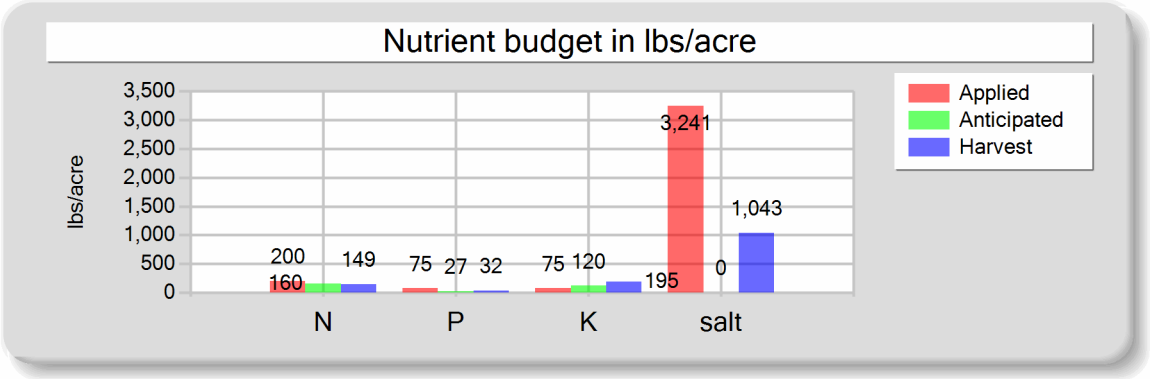
Fresh water applied	
12,352,500.00	*gallons*
454.90	*acre-inches*
28.43	*inches/acre*

Process wastewater applied	
1,476,000.00	*gallons*
54.36	*acre-inches*
3.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 10/28/2022: Triticale, soft dough

Field name: Field 5 Crop: Triticale, soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	192.91	74.90	74.90	2,767.88
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	473.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.91	74.90	74.90	3,241.04
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	149.21	32.48	194.54	1,042.91
Nutrient balance	50.71	42.42	-119.64	2,198.12
Applied to removed ratio	1.34	2.31	0.39	3.11

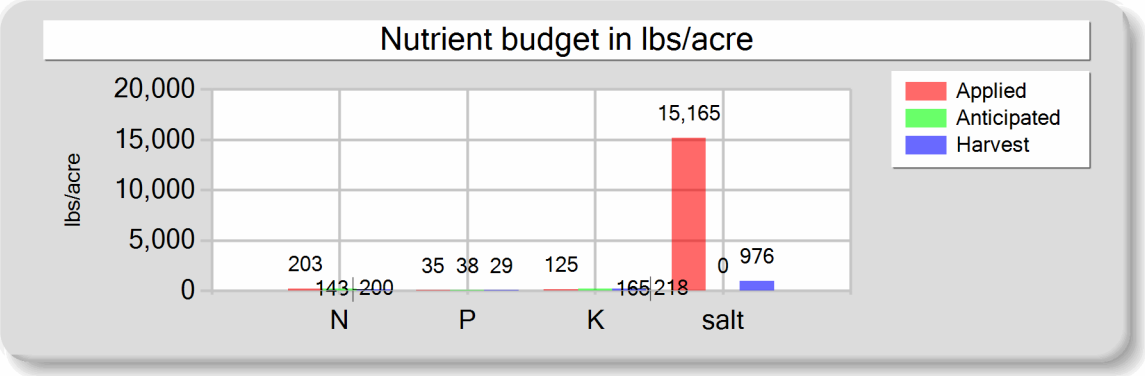
Fresh water applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
9.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 05/31/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	196.35	35.23	124.74	13,686.75
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,478.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.35	35.23	124.74	15,165.38
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	143.39	28.88	217.87	976.45
Nutrient balance	59.96	6.35	-93.13	14,188.93
Applied to removed ratio	1.42	1.22	0.57	15.53

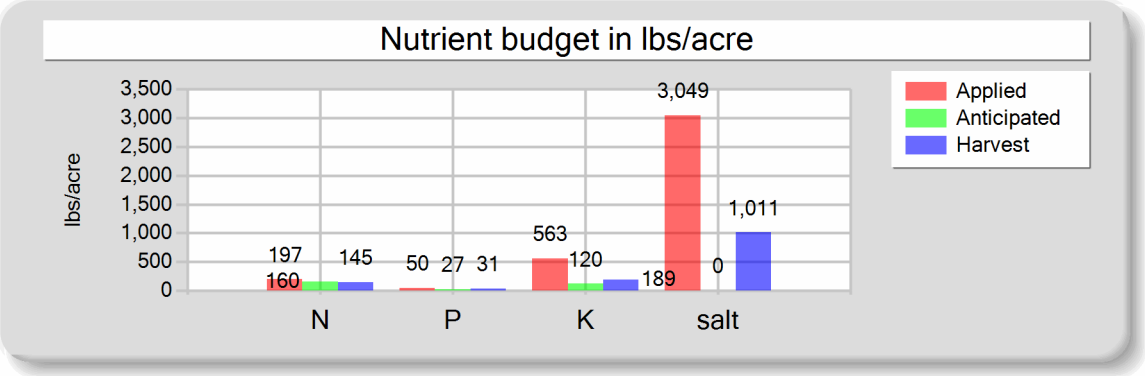
Fresh water applied	
10,125,000.00	*gallons*
372.87	*acre-inches*
31.07	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 10/29/2022: Triticale, soft dough

Field name: Field 6 Crop: Triticale, soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.11	49.51	563.30	2,604.94
Fresh water	0.00	0.00	0.00	443.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.11	49.51	563.30	3,048.53
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	144.65	31.49	188.60	1,011.06
Nutrient balance	52.46	18.02	374.70	2,037.47
Applied to removed ratio	1.36	1.57	2.99	3.02

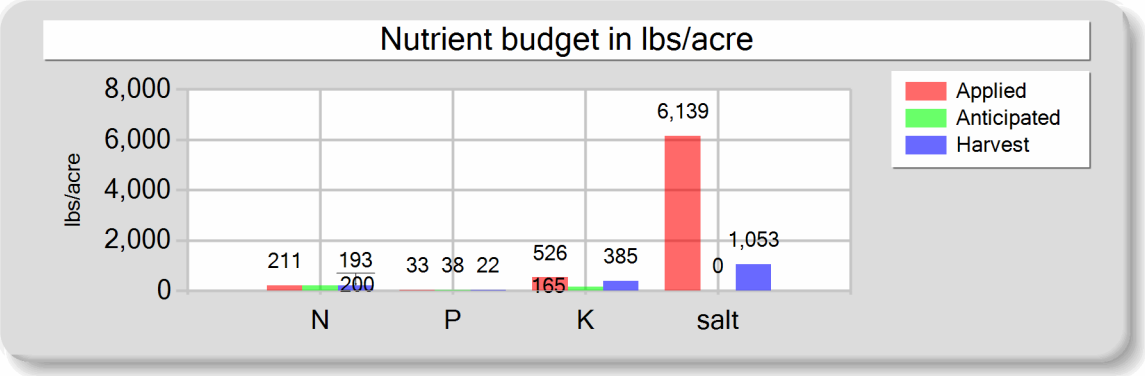
Fresh water applied	
10,125,000.00	*gallons*
372.87	*acre-inches*
9.32	*inches/acre*

Process wastewater applied	
2,952,000.00	*gallons*
108.71	*acre-inches*
2.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 05/29/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.41	33.26	525.61	4,737.02
Fresh water	0.00	0.00	0.00	1,401.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.41	33.26	525.61	6,138.76
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	192.70	22.09	384.93	1,052.80
Nutrient balance	18.71	11.17	140.68	5,085.96
Applied to removed ratio	1.10	1.51	1.37	5.83

Fresh water applied	
31,995,000.00	*gallons*
1,178.27	*acre-inches*
29.46	*inches/acre*

Process wastewater applied	
3,888,000.00	*gallons*
143.18	*acre-inches*
3.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Solids

Sample and source description: Separator Solids

Sample date: 11/15/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 81.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,000.00	8,930.00	8,930.00							33.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	2,080.00	6,120.00							47.10
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 05/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	1,830.00	6,480.00							71.10
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,000.00	8,420.00	10,600.00							35.90
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	280.00	148.00	0.00	0.00	71.10	761.00								5,830.00	3,820
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 03/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	322.00	182.00	0.00	0.00	84.70	986.00								5,840.00	4,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	252.00	143.00	0.00	0.00	41.00	648.00								5,550.00	5,840
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	252.00	213.00	0.00	0.00	31.50	541.00								6,150.00	8,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 12/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	750.00	448.00	0.00	0.00	59.80	1,050.00								9,430.00	9,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

TID Lateral 5

TID Lateral 5

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	210
DL	0.50	0.50	0.50								1.00	10

TID Lateral 5 1/2

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

TID Lateral 6

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

TID Lateral 7

TID Lateral 7

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	195
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 10/30/2022: Triticale, soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,380.00	810.00	6,140.00		9.76
DL	100.00	100.00	100.00		0.01

Field 1 - 05/28/2023: Corn, silage

Field 1 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,490.00	420.00	5,220.00		6.60
DL	100.00	100.00	100.00		0.01

Field 2 - 10/31/2022: Triticale, soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,680.00	1,080.00	7,050.00		8.19
DL	100.00	100.00	100.00		0.01

Field 2 - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,990.00	530.00	7,490.00		6.10
DL	100.00	100.00	100.00		0.01

Field 3 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,920.00	920.00	6,340.00		6.55
DL	0.50	0.50	0.50		0.01

Field 3 - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,170.00	490.00	6,160.00		6.60
DL	100.00	100.00	100.00		0.01

Field 4 - 10/29/2022: Triticale, soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,110.00	850.00	5,960.00		8.64
DL	100.00	100.00	100.00		0.01

Field 4 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,870.00	570.00	6,160.00		7.40
DL	100.00	100.00	100.00		0.01

Field 5 - 10/28/2022: Triticale, soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	441.00	960.00	5,750.00		6.85
DL	100.00	100.00	100.00		0.01

Field 5 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,830.00	570.00	4,300.00		6.60
DL	100.00	100.00	100.00		0.01

Field 6 - 10/29/2022: Triticale, soft dough

Forage

Sample and source description: Forage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	470.00	8,190.00		7.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

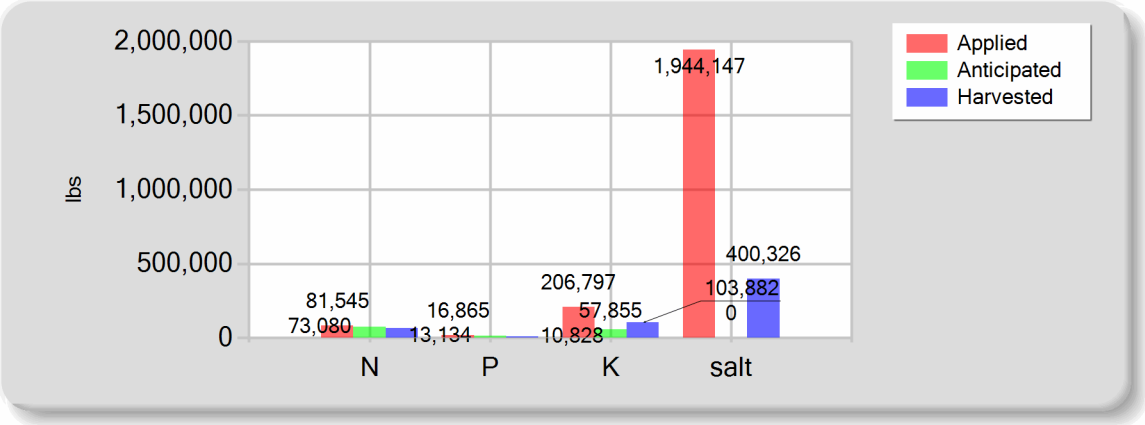
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

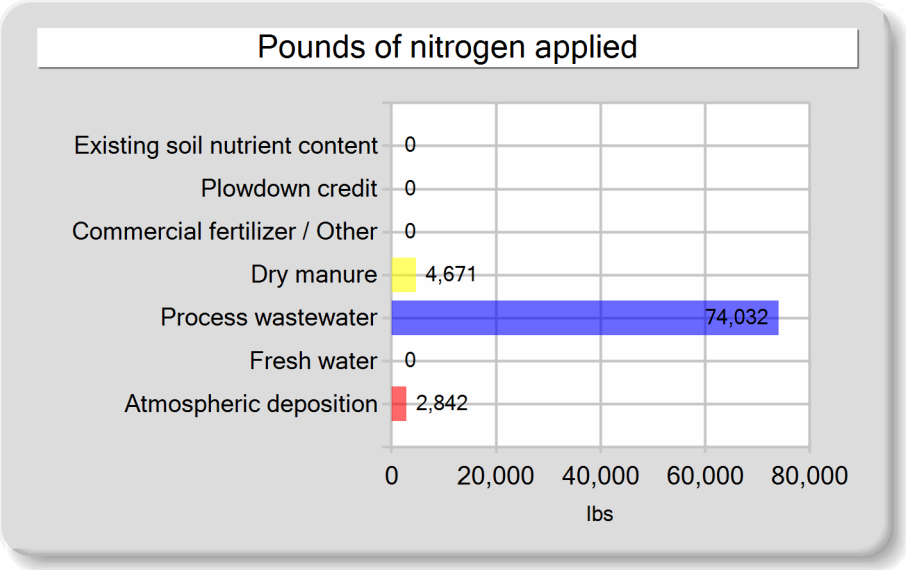
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

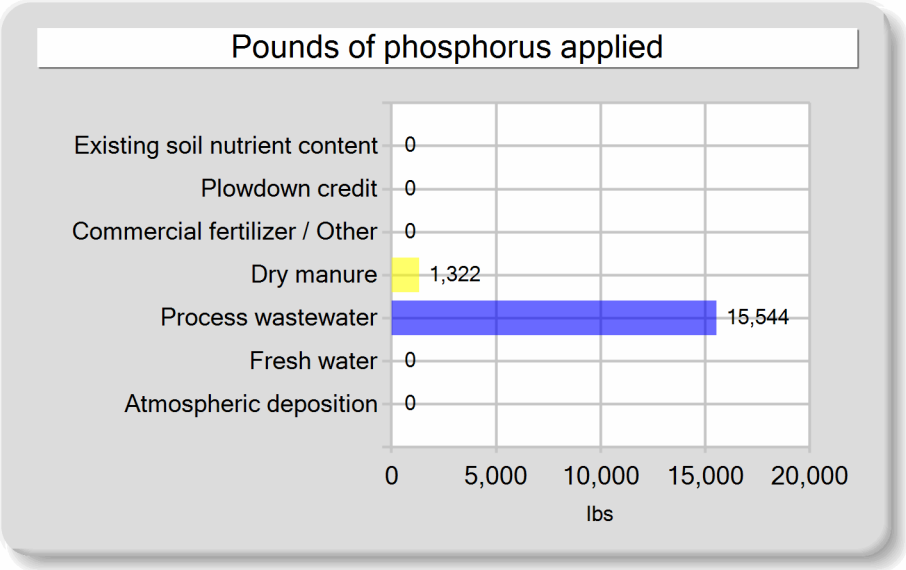
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	4,671.15	1,321.53	2,395.68	197,455.50
Process wastewater	74,031.90	15,543.73	204,401.56	1,376,560.49
Fresh water	0.00	0.00	0.00	370,130.58
Atmospheric deposition	2,842.00	0.00	0.00	0.00
Total nutrients applied	81,545.05	16,865.27	206,797.25	1,944,146.57
Anticipated crop nutrient removal	73,080.00	13,134.10	57,855.00	0.00
Actual crop nutrient removal	67,105.92	10,827.98	103,881.88	400,325.99
Nutrient balance	14,439.13	6,037.29	102,915.37	1,543,820.58
Applied to removed ratio	1.22	1.56	1.99	4.86

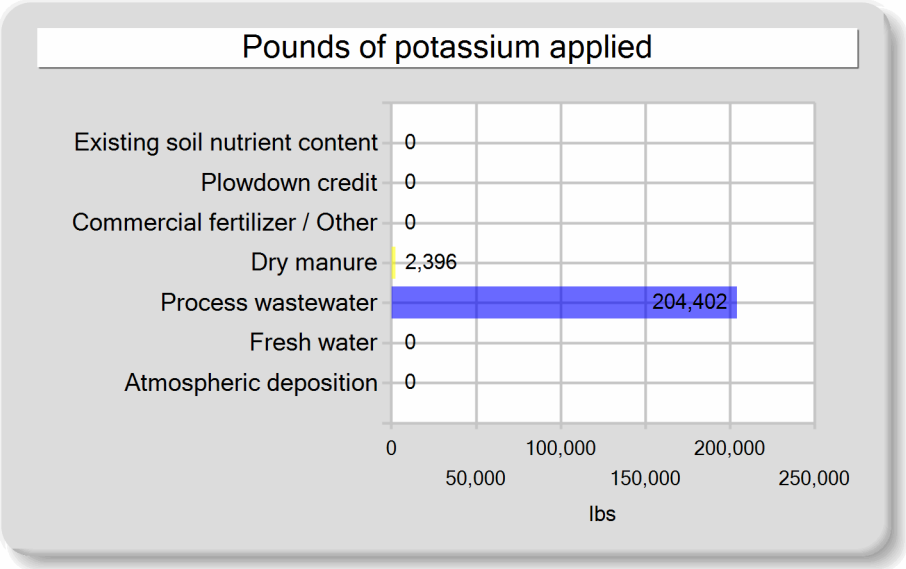
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

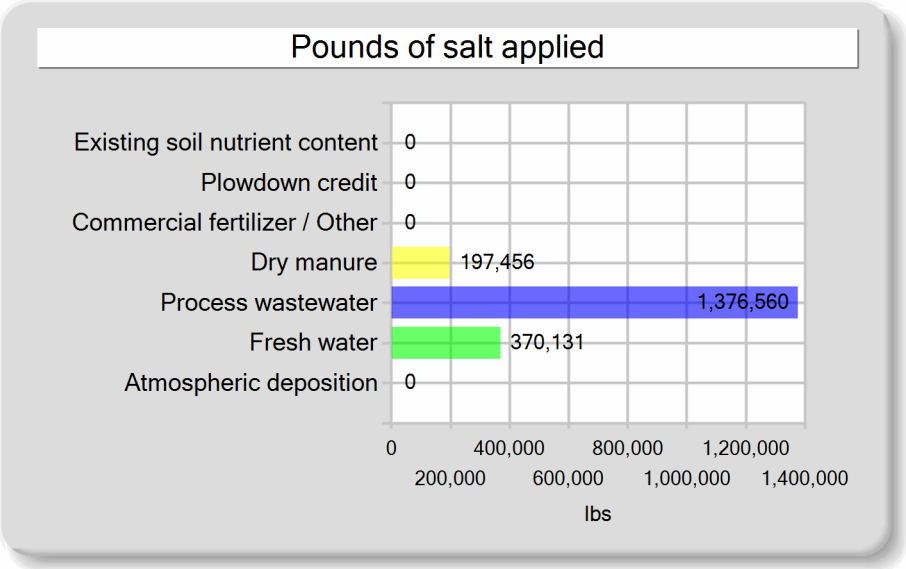


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Michael Carvalho (Jun 24, 2024 16:39 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Michael Carvalho	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 24, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Carvalho Farms
11433 Crows Landing Rd.
Crows Landing, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T3346024A	Electrical Conductivity (EC)	EPA 120.1 and	3.56	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.199	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2280	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Grower

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T3346024B	Electrical Conductivity (EC)	EPA 120.1 and	3.55	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.154	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2270	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

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Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 3	T3346024C	Electrical Conductivity (EC)	EPA 120.1 and	1.41	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	902	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

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Grower

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F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 4	T3346024D	Electrical Conductivity (EC)	EPA 120.1 and	3.57	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2280	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	93.0	98.2 (T3346022G)	100 (T3346022G)	1.81
Electrical Conductivity (EC)	99.6			0.253
Nitrate Nitrogen (NO3-N)	102	98.4 (T3348021A)	103 (T3348021A)	4.70

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Compliance Analysis Report

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 12:00 PM
SUBMITTED BY: KK
ANALYZED DATE: 12/15/23

Process Waste Water (H3)
SOURCE: Lagoon Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units	Lbs/10000 Gal as Rec'd
WWSQ4	T3346024E	Total Dissolved Solids (TDS)	SM 2540 C-1997	9560	mg/L	797
		Total Phosphorus (TP)	EPA 200.7	59.8	mg/L	4.99
		Electrical Conductivity (EC)	EPA 120.1 and	9.43	mmhos/cm	N/A
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-	448	mg/L	37.4
		Total Kjeldahl Nitrogen (TKN)	SM 4500-NH3 C-	750	mg/L	62.6
		Total Potassium (TK)	EPA 200.7	1050	mg/L	87.8
		Soluble Salts (SALT-SOL)	Calculation	6030	mg/L	503

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.2			252 (T3346018A)	244 (null)	3.39
Total Dissolved Solids (TDS)	99.1			4700 (T3352013A)	4440 (Dup)	5.69
Total Kjeldahl Nitrogen (TKN)	105			392 (T3346018A)	395 (null)	0.712
Total Phosphorus (TP)	101			29.0 (T3346011A)	24.6 (DUP)	16.4
Total Potassium (TK)	105			730 (T3346011A)	689 (DUP)	5.80

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DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Carvalho
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By:
Project ID: K.Jones

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/12	10:16	Domestic Wells	DW 1	16.0°C		X				
2	12/12	10:22	Domestic Wells	DW 2	16.0°C		X				
3	12/12	10:28	Domestic Wells	DW 3	15.6°C		X				
4	12/12	10:36	Domestic Wells	DW 4	15.0°C		X				
5	12/12	10:13	WASTEWATER	WWSOH	17.0°C			Y			
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged In By:	LABORATORY NUMBER
	KK	F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	12-12-23 12:00			T3346024

Carvalho 2023 Annual Report

Final Audit Report 2024-06-24

Created:	2024-06-24
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
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